

## Automation Framework Master Notes

### 1. Page Object Model (POM)

- Separates locators, actions, and test logic.
- Benefits: cleaner code, easier maintenance, reusable methods.
- Example structure: pages/, tests/, utilities/

### 2. Reusable Utilities

- Create utility classes for screenshots, waits, scrolling, clicking, reports, and browser setup.
- Example: `WaitUtils.waitForElement()`, `Screenshot.takeScreenshot()`

### 3. Data-Driven Testing

- Use TestNG DataProvider, Excel, CSV, and JSON instead of hardcoded test data.

### 4. Cross-Browser Support

- Framework should support Chrome, Firefox, and Edge.
- Example: `<parameter name="browser" value="chrome"/>`

### 5. Proper Reporting

- Generate screenshots, HTML reports, Word/PDF reports, and logs.
- Use tools like Extent Reports, Allure Reports, and Apache POI.

### 6. CI/CD Integration

- Integrate with Jenkins, GitHub Actions, and Azure DevOps.
- Automate nightly regression, smoke tests, and deployment validation.

### 7. Logging

- Avoid `System.out.println()`.
- Use Log4j or SLF4J for cleaner debugging and professional logs.

### 8. Explicit Waits

- Avoid `Thread.sleep()`.
- Use `WebDriverWait` for stable synchronization.

### 9. Centralized Configuration

- Store URLs, passwords, browsers, and environments in `config.properties`.

### 10. Exception Handling

- Capture failures, take screenshots automatically, and produce readable messages.

### **11. Modular Design**

- Separate tests, pages, utilities, reports, data, and locators.

### **12. Parallel Execution**

- Example: parallel="tests" thread-count="3"
- Improves regression execution speed.

### **13. API + UI Hybrid Framework**

- Combine Selenium + REST Assured + Database validation.
- Enterprise frameworks commonly support UI + API + DB testing.

### **14. Retry Mechanism**

- Automatically rerun flaky tests.

### **15. Screenshot on Failure**

- Capture screenshots automatically when tests fail.

### **16. Environment Support**

- Support QA, UAT, Production, and Staging environments.

### **17. Version Control**

- Use Git and GitHub for collaboration and CI/CD integration.

### **18. Clean Naming Conventions**

- Examples: LoginPage, CartPage, WaitUtils, BaseTest.

### **Enterprise-Level Stack**

- Java
- Selenium
- TestNG
- Maven
- Jenkins
- GitHub
- REST Assured
- Extent Reports
- Apache POI
- Log4j
- Docker